

Junyu (James) Guo

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EDUCATION

- **University of California, Berkeley** Berkeley, CA
Ph.D. Candidate in Industrial Engineering and Operations Research *Aug 2024 – May 2029 (expected)*
 - **Advisor:** Prof. Javad Lavaei. Research on LLM agents and reinforcement learning
 - **Selected Coursework:** Advanced Natural Language Processing, Optimization
- **Tsinghua University** Beijing, China
B.S. in Mathematics; GPA: 3.90/4.00; Rank: 3/55 *Sep 2020 – Jun 2024*
- **Cornell University** Ithaca, NY
Exchange Student; GPA: 4.13/4.30 *Jan 2023 – May 2023*

INDUSTRY EXPERIENCE

- **TikTok, Data and Privacy Office** San Jose, CA
Machine Learning Engineer Intern *May – Aug 2026*
 - Built the agent harness for AI root-cause analysis of bug tickets used by R&D engineers: parallel retrieval over design docs, configs, logs, and past tickets, with evidence cross-validation and confidence-gated verdicts.
 - Designed its self-improvement loop: recurring evaluation on closed tickets attributes each failure to retrieval, guardrails, routing, or knowledge and proposes code fixes for review; time to close invalid defects fell 39%.
 - Developed a risk-discovery agent for pre-launch review that reads product requirement docs and the corresponding codebase to proactively detect privacy and compliance risks in new features, at 30% lower latency.

RESEARCH EXPERIENCE

- **Context-Update Failures in LLM Agents** Preprint, Sep 2026
With Prof. Javad Lavaei, Prof. Costas Spanos, and Prof. James Demmel *Berkeley, CA*
 - Identified *stale binding*: LLM agents answer with an outdated value of a variable even though the updated value is still in context. Showed that frontier reasoning models also fail to recover the current state.
 - Built Controlled In-Context Memory (CICM), a benchmark that measures how well models track and use updated information in conversations and agent logs.
 - Showed with probes that models still encode the updated value, and traced the failure to *attention drift* toward old values, with a mathematical account in a one-layer transformer.
 - Designed an inference-time attention intervention that needs no further training and corrects most old-value errors across model families while preserving nearly all initially correct answers.
- **Training LLMs to Express Uncertainty Explicitly** Preprint, Apr 2026
With Prof. Javad Lavaei, Prof. Costas Spanos, Prof. Ming Jin, and Dr. Shangding Gu *Berkeley, CA*
 - Post-trained LLMs across various model families with reinforcement learning so that the model states its own uncertainty: a verbalized confidence score after reasoning, or an `<uncertain>` marker during reasoning.
 - Improved answer quality in adaptive retrieval-augmented generation while calling external tools far less often, by letting the model's own uncertainty decide when retrieval is needed.
 - Interpreted the internal mechanisms behind both signals with logit-lens, representation-similarity (CKA), and activation-patching analyses.
 - Found that verbalized confidence sharpens a confidence structure already present in the pretrained model, while the marker teaches the model to flag high-risk reasoning steps.
- **Diffusion-Regularized Constrained Offline Reinforcement Learning** NeurIPS 2025
With Prof. Javad Lavaei and Prof. Ming Jin *Oct 2024 – Jun 2025*
 - Developed a constrained offline RL method that improves task performance while satisfying safety constraints, with no additional interaction with the environment.
 - Modeled the behavior distribution of the offline dataset with a diffusion policy and used its score function to regularize policy optimization, which reduces distribution shift toward actions unsupported by the data.
 - Designed a gradient-based policy update balancing reward, safety, and closeness to the data; achieved higher returns and fewer constraint violations than baselines on Safety-Gymnasium and Bullet-Safety-Gym.

PUBLICATIONS & PREPRINTS

- **Groundability, Not Scale Alone: When Weak Reviewers Can Audit Strong Coding Agents:** Preprint, 2026. [\[Paper\]](#)
- **When Context Changes: Understanding Update Failures in LLMs:** Preprint, 2026. [\[Paper\]](#)
- **LLMs Should Express Uncertainty Explicitly:** Preprint, 2026. [\[Paper\]](#)
- **Don't Trade Off Safety: Diffusion Regularization for Constrained Offline RL:** NeurIPS 2025. [\[Paper\]](#)
- **Meta Thinker: Thinking What AI Thinks:** NeurIPS 2025 MATH-AI Workshop. [\[Paper\]](#)
- **StyleBench: Evaluating Thinking Styles in Large Language Models:** Preprint, 2025. [\[Paper\]](#)
- **Reinforcement Learning for SBM Graphon Games with Re-Sampling:** Preprint, 2023. [\[Paper\]](#)

SKILLS

- **Programming:** Python, C++, R, MATLAB, Unix shell, $\LaTeX$
- **LLMs and Agents:** Post-training with reinforcement learning, evaluation and benchmark design, agentic workflows, retrieval-augmented generation
- **Interpretability:** Probing, logit lens, activation patching, attention analysis
- **Reinforcement Learning:** Offline and safe RL, diffusion policies, constrained policy optimization

TEACHING AND SERVICE

- **Graduate Student Instructor:** INDENG 160, Nonlinear and Discrete Optimization, UC Berkeley (Fall 2025, Spring 2026)
- **Reviewer:** ICLR 2027, AAAI 2027, NeurIPS 2026, ICML 2026, COLM 2026

HONORS AND AWARDS

- Wolff Fellowship, UC Berkeley (2024)
- First Prize, China Undergraduate Mathematics Competition (2021)
- Scholarship for Merit Students, Tsinghua University (2020, 2021, 2022); Mathematics Department Academic Scholarship (2021)
- First Prize, China High School National Mathematics Olympic League (2018, 2019)